

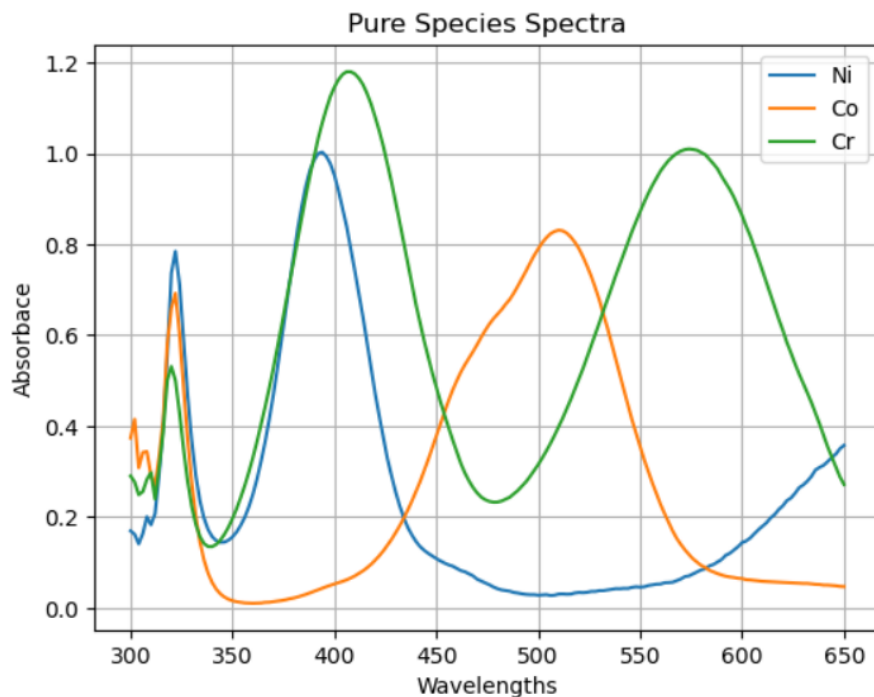
CH5440 ASSIGNMENT 4 : PCR

[ME20B072 GIRISH MADHAVAN]

- a. Using the pure component spectra, identify the wavelengths at which the pure species have maximum absorbance. Denote them as $\lambda_{\max}$ values and report them. Build a calibration model using OLS by selecting the mixture absorbances at the $\lambda_{\max}$ wavelengths. Report the RMSE obtained using LOOCV.

SOLN :

- Inorder to compute the calibration matrix we construct the data matrix by using sample absorbances corresponding to the $\lambda_{\max}$ values and then perform OLS regression between the data matrix and concentration matrix.
- Also to incorporate LOOCV we iterate within all the 26 mixtures present in the data matrix, each time leaving one of the rows in the matrix to be predicted using OLS.
- This Predicted solution is used to find the error from the actual value which is then squared and added with the error from the previous iteration.
- This total error is then divided by the number of mixtures (26) and its square root is taken to get RMSE



- Regression Equation : Here we don't do mean shifting because we try to get a model that passes through the origin

$$C_{N \times p} = A_{N \times p} B_{p \times p}$$

$$B = (A^T A)^{-1} A^T C$$

C is the Concentration matrix

A is the Data matrix

B is the Calibration matrix

- Now we have two cases for obtaining the absorbances at λ_{max} values because the maximum absorbances of Cr and Ni are close to each other which would make identification of these species quite difficult
- Fortunately Cr has two peaks in its absorbance graph, so case 1 would consider the λ_{max} values to be close to each other for Cr and Ni and case 2 would consider the other peak of Cr

CASE 1 :

Maximum Wavelengths (nm)	Absorbance of Pure Species
510	0.83097
406	1.179606
394	1.002746
RMSE	0.0025980526

CASE 2 :

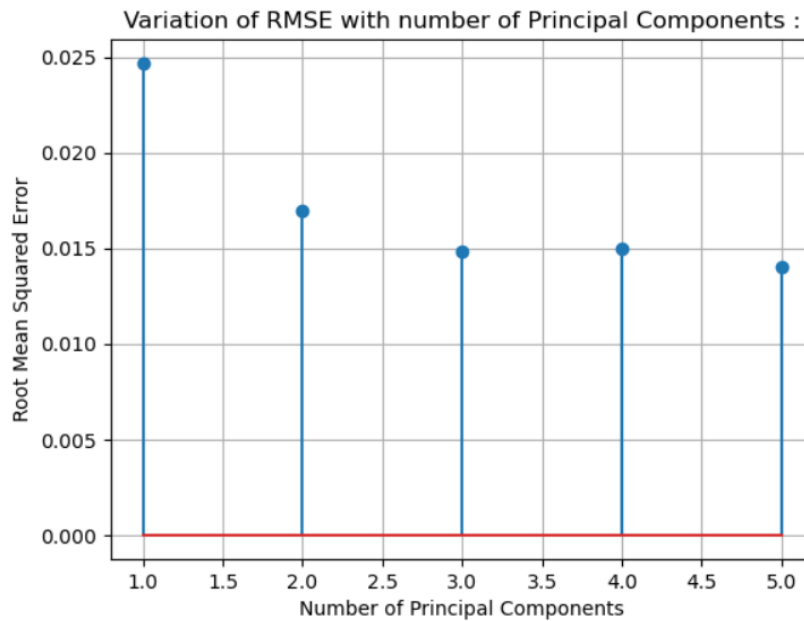
Maximum Wavelengths (nm)	Absorbance of Pure Species
510	0.83097
574	1.009735
394	1.002746
RMSE	0.0008110347

- The **RMSE** has **decreased** by using the maximum wavelength corresponding to the other peak of **Cr**

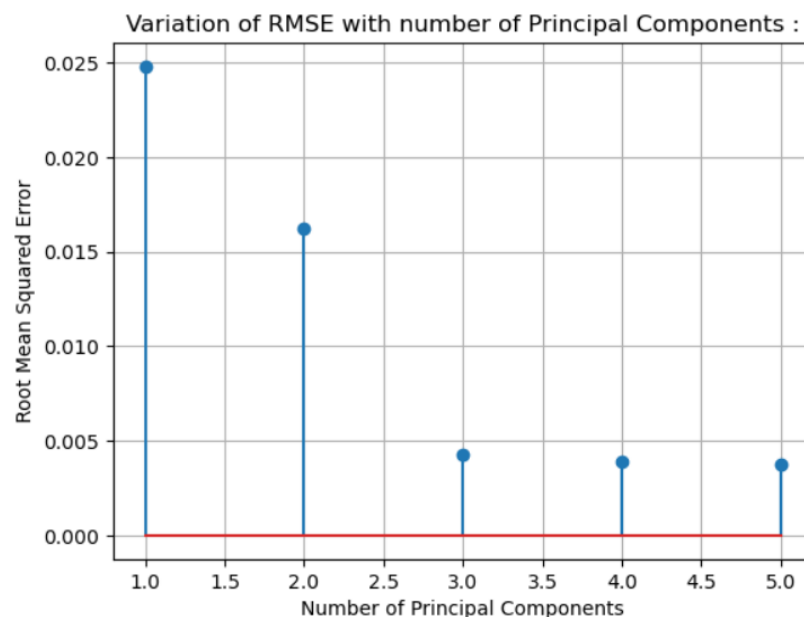
- b. Develop a multivariate calibration model using PCR. Evaluate the RMSE for different choices of number of PCs (from 1 to 5) selected in step 1 of PCR. Report the LOOCV RMSE results in the form of a table for different number of PCs chosen. Are you able to estimate the number of species correctly using RMSE?

SOLN : Again **two cases exist** here, because of ambiguity in way the data matrix is built :

CASE 1 : The data matrix is comprised of the **first samples of the 26 mixtures alone**



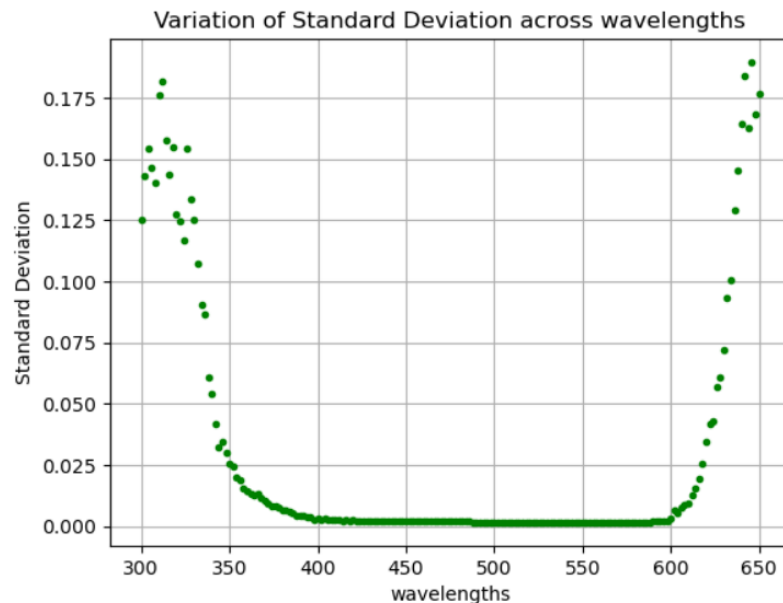
CASE 2 : The data matrix is comprised of the **mean of all the samples** corresponding to each mixture, resulting in a **26 x 176 matrix**



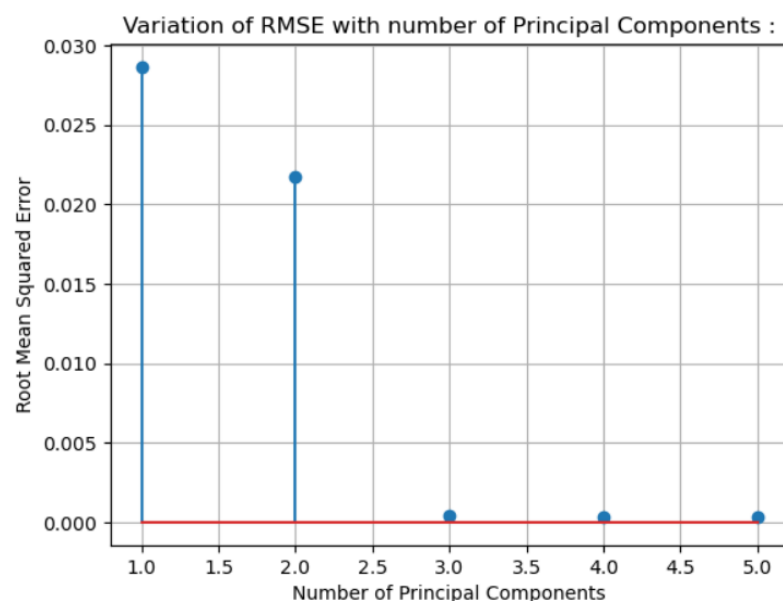
RMSE values in CASE 2 are lower than that of CASE 1, so taking the **mean of readings is better**. Also, the **RMSE has a sharp decrease till 3 PCs** and it **remains almost constant** after that. This suggests that there are **3 species present in the mixture**.

- c. Estimate the standard deviation of errors in absorbance measurements using the five replicates for each wavelength and for each mixture. Assume that the error standard deviations vary significantly with respect to wavelength but are almost same for all mixtures. Use these standard deviations to scale the absorbance measurements for each wavelength before applying PCA in the first step for different choices of PCs (1 to 5) chosen. Determine the RMSE using LOOCV. Do your results improve? Are you able to correctly determine the number of species using LOOCV?

SOLN: The below plot shows the variation of standard deviation across the wavelengths assuming the variances across samples to be constant [here the mean of variances for each wavelength is taken final value]



- Now we use the **1 x 176 matrix of standard deviations** to obtain the **diagonal cholesky matrix** which will be used for **scaling the data matrix**.
- After this step we perform **Principal Component Regression** again and compute the **RMSE** for **varying number of principal components**



- Again we see a **Sharp decrease in RMSE till 3 PCs** and **it remains constant**, this would suggest that there are **3 species in the mixtures**